8 **(a)** Fig. 8.1 shows a wire carrying a large current.

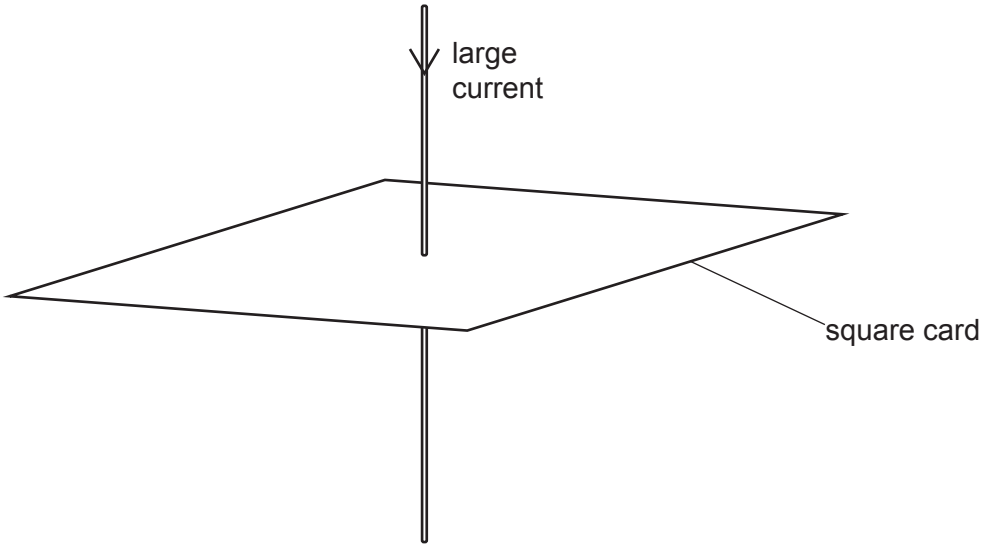


Fig. 8.1

(i) Fig. 8.2 shows the square card viewed from above.

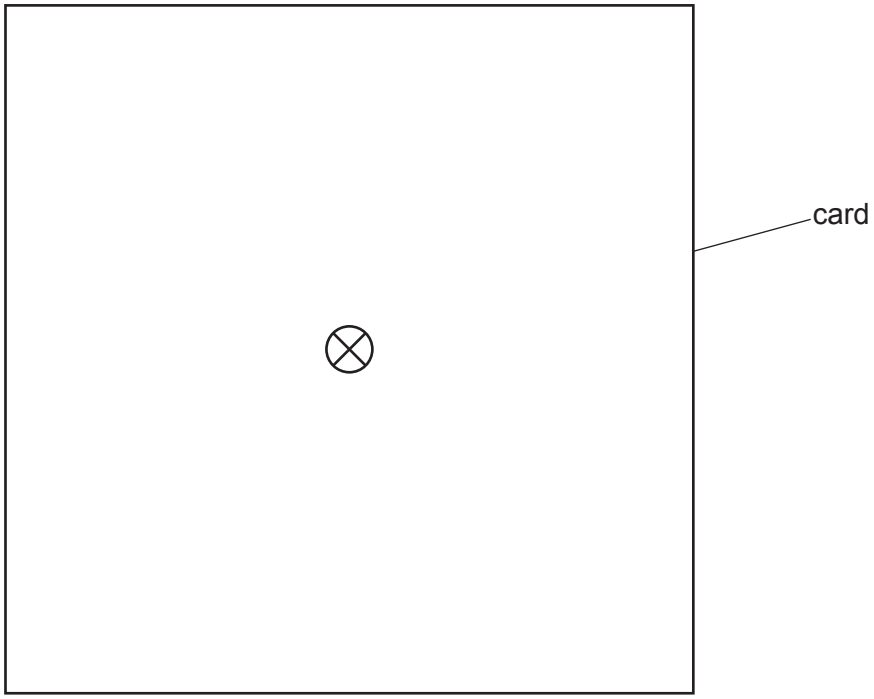


Fig. 8.2

On Fig. 8.2, draw **three** magnetic field lines that indicate the direction of the magnetic field and how its strength varies with distance from the wire. [3]

(ii) The current in the wire increases and the direction of the current is reversed.

State how these changes affect the magnetic field.

.....

..... [2]

(b) Electricity is transmitted at high voltage.

Explain why a high voltage increases the efficiency of transmission even with thinner wires.

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.....

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..... [3]

[Total: 8]